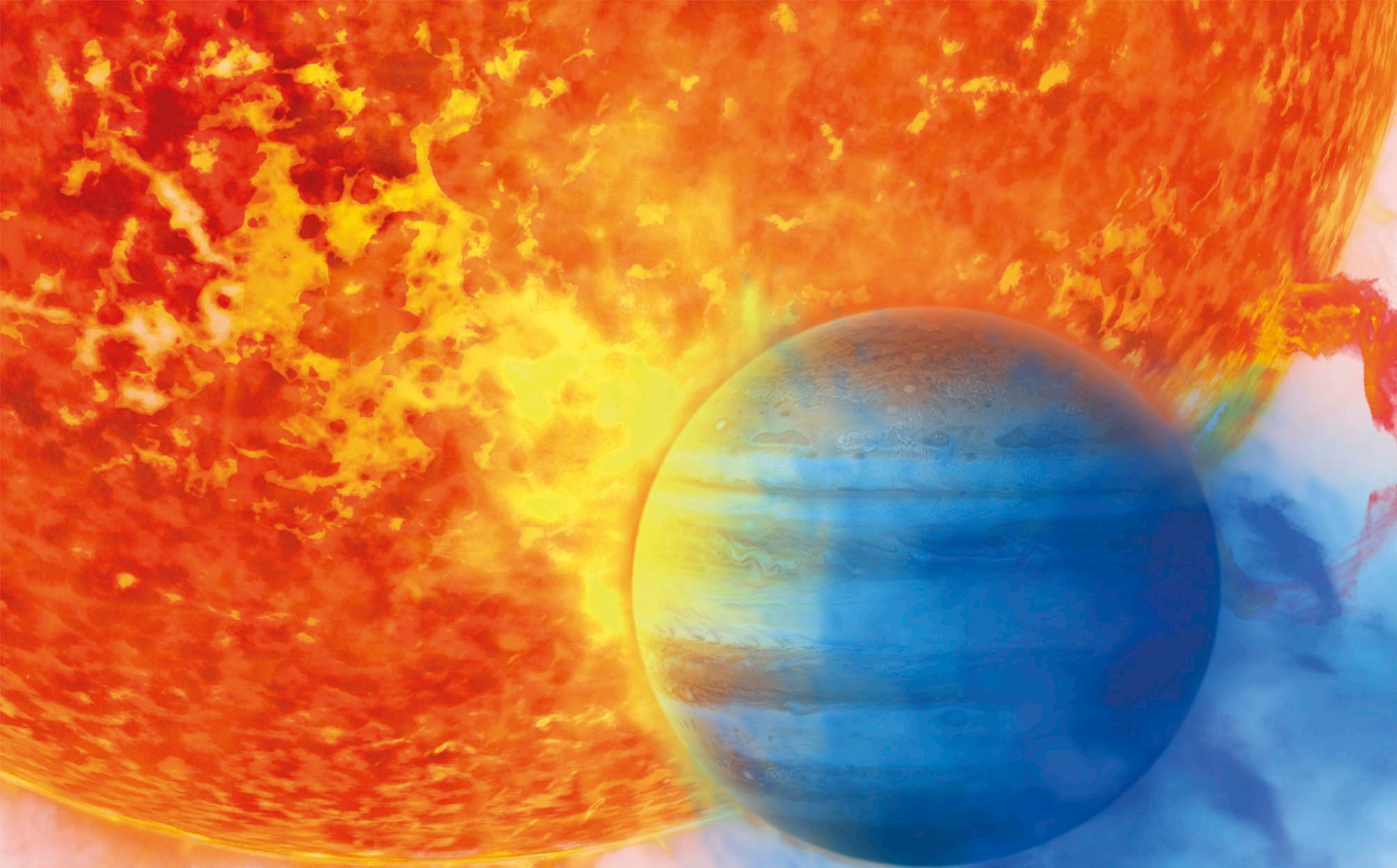
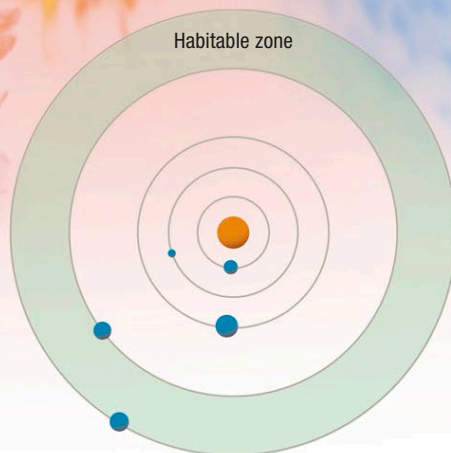




#### ▷ **Kepler-62 system**

In 2013, the Kepler Space Telescope discovered five planets orbiting the star Kepler-62, which lies 1,200 light-years from Earth. Two of these planets orbit in an area known as the habitable zone (or “Goldilocks zone”), where temperatures are just right for water to exist at the surface.



#### △ **"Hot Jupiters"**

An exoplanet of this type orbits its host star at a distance of less than 46 million miles (75 million km), which is much closer than Jupiter orbits the Sun. It is scorched by its host star, producing extreme weather in its atmosphere.

### **Properties of Exoplanetary Systems**

More than half of known exoplanetary systems consist of a single star with a single planet orbiting it (in many of these there may be other, so far undetected, planets). However, as of February 2016, more than 500 multiplanetary systems—containing two or more planets—had been discovered. Some contain five, six, or in a few cases, seven planets. Of particular interest in any planetary system is its habitable zone. This is the region around the central star where temperatures are right for water—essential for life as we know it—to collect on the surface of any planet with a rocky surface. A planet that looks like it could be rocky and is in a star’s habitable zone is of extra interest because it could harbor life.

#### ▽ **Kepler-62 planets**

An artist’s impression of the five Kepler-62 planets is shown below: they are too far away to photograph. The two on the right may be rocky planets with surface water. Little is known about the others except for their size and the fact that their surface must be extremely hot.

